perplexity

Step 2: How regression testing works

Regression testing works by keeping a set of repeatable checks for important features, then running those checks again whenever the software changes. If a previously passing test now fails, the team investigates whether the recent change caused the problem.[1] [2]

Test case

A **test case** is one specific instruction for checking software behavior. It usually has:

- **Title:** What is being checked
- **Steps:** What the tester or automation does
- **Input:** Data used, such as a username and password
- **Expected result:** What should happen
- **Actual result:** What actually happened

Example:

Field	Example
Title	Verify a user can log in with valid credentials
Steps	Open the login page, enter a valid email and password, click “Log in”
Expected result	User reaches the dashboard
Regression failure	User stays on the login page or sees an error

A regression suite is made from many such test cases covering core features such as login, search, checkout, and payments.[2] [3]

Regression test suite

A **regression suite** is a curated collection of tests that protects the software’s existing behavior. Teams do not necessarily rerun every possible test every time; they often use a risk-based approach, prioritizing critical user journeys and areas affected by the change.[4] [2]

Common layers include:

- **Smoke tests:** A small, fast set checking whether the app’s most basic functions work.
- **Targeted regression tests:** Tests focused on the feature or module that changed.
- **Full regression suite:** A broader set run before major releases or significant architectural changes.[5] [2]

When teams run it

Regression testing is typically run after changes that could affect existing behavior, including new features, bug fixes, code refactoring, dependency updates, and configuration changes. It is also a standard gate before deploying to production.[6] [1]

Trigger	Typical testing scope
Developer submits a pull request	Fast, targeted tests for affected areas
Code is merged	Core regression and smoke tests
Bug is fixed	Retest the fix plus related regression tests
Major release is planned	Full regression suite
Production hotfix is deployed	Critical smoke and focused regression checks [2] [7] [8]

Manual vs automated

Approach	How it works	Best for
Manual regression testing	A person follows test steps and compares actual results with expected results	New or unstable features, visual design, usability, and exploratory checks
Automated regression testing	Scripts run the checks automatically, often in CI/CD	Stable, repetitive, high-value tests that must run frequently [2] [5]

In modern teams, automated regression tests often run in a **CI/CD pipeline**: when a developer pushes code, the system builds the app and runs selected tests automatically. A failed test can block the merge or deployment, giving fast feedback before problems reach users.[8] [1]

Step 3 preview

Next, I'll explain **Step 3: How to perform regression testing**, with a beginner-friendly workflow:

1. Identify what changed
2. Choose which tests to run
3. Prepare the test environment and data
4. Execute the tests
5. Record results and report failures
6. Fix, retest, and update the suite

Shall I continue with **Step 3**?

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